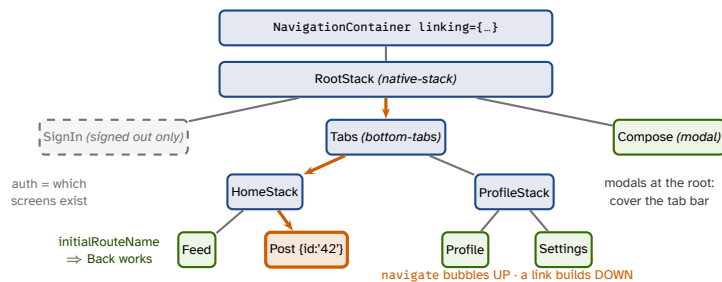


**In one line:** Navigation is a **tree of navigators** holding one serialisable **state object**; actions (navigate, goBack) **bubble up** to the nearest navigator that can handle them, deep links are **parsed down** into that state, and screens **stay mounted** under a push — so “appear” is focus, not mount.



where `https://ex.com/post/42` lands

- 1 OS: iOS universal link / Android App Link  
cold: `Linking.getInitialURL()` · warm: 'url' event
  - 2 strip a prefixes entry → `post/42`
  - 3 match config: `Post: 'post/:id'`
  - 4 build state: `Root → Tabs → HomeStack`  
`routes [Feed, Post{id:'42'}] → reset`
- signed out ⇒ Post not rendered ⇒ link dropped: stash, replay

## Navigators — what they are natively

- native-stack** — via `react-native-screens`: a real **UINavigationController** / Android **Fragment**. Native push, swipe-back, large titles. Default choice; custom transitions limited.
- stack** — JS: Animated + gesture-handler. Any transition you can draw; costs JS/UI work, never quite UIKit.
- bottom-tabs** — JS tab bar, screens **lazy**, then **kept mounted**.
- drawer** — reanimated + gesture-handler. `freezeOnBlur` stops re-rendering hidden screens.
- Nesting rules**. Each navigator owns its own history. `navigate(name)` is tried by the current navigator, then its parents — never *into* an unrendered child: use `navigate('Tabs', {screen, params})`. `goBack` pops the nearest navigator with history. `setOptions` affects only the screen's own navigator. Params do **not** flow to child screens.
- Shape decides UX**: tabs *inside* a stack screen → pushed detail hides the tab bar; a stack *inside* each tab → tab bar stays, per-tab history (the UIKit `UITabBarController-of-navs` pattern).
- Lifecycle ≠ component lifecycle**. Push B over A: A is *not* unmounted; Back unmounts B. So `useEffect = viewDidLoad`; `useFocusEffect = viewWillAppear/Disappear`; `useIsFocused` re-renders; listeners `focus · blur · beforeRemove` (unsaved-changes guard).

## Example — typed params, linking, focus

```
type HomeParams = { Feed: undefined; Post: { id: string } };
type TabParams = { HomeTab: NavigatorScreenParams<HomeParams> };
type RootParams = { Tabs: NavigatorScreenParams<TabParams>;
  SignIn: undefined; Compose: { draft?: string } };
type PostProps = NativeStackScreenProps<HomeParams, 'Post'>;
declare global { namespace ReactNavigation { // typed hooks
  interface RootParamList extends RootParams {} } }
const linking: LinkingOptions<RootParams> = {
  prefixes: ['myapp://', 'https://ex.com'],
  config: { screens: { Compose: 'compose', Tabs: { screens: {
    HomeTab: { initialRouteName: 'Feed', // Feed under Post
    screens: { Feed: 'feed', Post: 'post/:id' } } } } } } };
useFocusEffect(useCallback(() => { // runs on every focus
  const sub = feed.subscribe(); return () => sub.remove(); }, []));
```

## Deep links — the native half

- iOS**: custom scheme in `CFBundleURLTypes`; universal links need the **Associated Domains** entitlement `applinks:ex.com` + `/well-known/apple-app-site-association`. Bare RN: AppDelegate forwards open url and continue `userActivity` to `RCTLinkingManager` (Expo wires it).
- Android**: `intent-filter VIEW + BROWSABLE`, `autoVerify="true"` + `/well-known/assetlinks.json`.
- Notification taps: override `getInitialURL/subscribe` — a tap is just a URL. Test: `xcrun simctl openurl booted <url> · adb shell am start -a android.intent.action.VIEW -d <url>`.

## Auth flows · persistence

- Auth = conditional screens**, not redirects: `{user ? <Screen Tabs /> : <Screen SignIn />}`. State flips → the tree swaps, `SignIn` leaves history (no Back to login), no navigate race.
- Persistence**: `initialState` + `onStateChange` → `AsyncStorage`, restored before first render. **Serialisable params** only (no callbacks); skip when opened by a link.

## Expo Router vs React Navigation

|        | React Navigation               | Expo Router (built on it)                                 |
|--------|--------------------------------|-----------------------------------------------------------|
| routes | components in code             | files: <code>[id].tsx</code> (group) <code>_layout</code> |
| links  | config by hand                 | automatic, one URL per file                               |
| types  | <code>ParamList</code> by hand | generated typed hrefs                                     |
| pick   | dynamic trees, non-Expo        | Expo, web, link-heavy apps                                |

## Traps

- ! Refetch in `useEffect` skips Back: `useFocusEffect(useCallback(...))`.
- ! `navigate('Home')` after login — use conditional screens.
- ! Stack in stack/tab = **two headers**: `headerShown: false`.
- ! Nested link without `initialRouteName` → no Back.
- ! push = new instance (`Profile→Profile`); navigate reuses.

## Remember

**Actions bubble up, links build down, pushes don't unmount.**